EE2080: Random Processes

Review: Probability and Random Variables

Anjana A. M.
Department of EE
Email: anjana.am@ee.iith.ac.in

July 28, 2025

Probability Space

Probability Theory deals with modeling

- **Random experiments**: the results are not deterministic, and their
- **Outcomes** : the possible results of the experiment.

A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

- **Sample space**: the set of all possible outcomes, denoted Ω
- **Sigma-algebra** of subsets of Ω - A collection $\mathcal{F}$ of subsets of Ω s.t.
  - E1. $\Omega \in \mathcal{F}$
  - E2. Closed under complements: $A \in \mathcal{F} \implies A^c \in \mathcal{F}$
  - E3. Closed under countable unions: $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

Probability Space: Continued

- The pair $(\Omega, \mathcal{F})$: **Measurable Space**

- **Probability measure** $\mathbb{P} : \mathcal{F} \to [0, 1]$

  P1. Non-negativity: For any $A \in \mathcal{F} : \mathbb{P}(A) \geq 0$

  P2. Normalization: $\mathbb{P}(\Omega) = 1$

  P3. Countable additivity: If A_i are disjoint, $\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i)$

Example

Consider the experiment of rolling a [redacted] die.

- Sample Space $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Sigma-algebra:
  - $\mathcal{F}_1 = \{\emptyset, \Omega\}$
  - $\mathcal{F}_2 = 2^{\Omega}$
  - $\mathcal{F}_3 = \{\emptyset, \{1, 2\}, \{3, 4, 5, 6\}, \Omega\}$
  - $\mathcal{F}_4 = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, \{3, 4, 5, 6\}, \{1, 3, 4, 5, 6\}, \{2, 3, 4, 5, 6\}, \Omega\}$
- Consider $\mathcal{F}_3 = \{\emptyset, \{1, 2\}, \{3, 4, 5, 6\}, \Omega\}$
  - $\mathbb{P}_1 = \{0, \frac{1}{3}, \frac{2}{3}, 1\}$
  - $\mathbb{P}_2 = \{0, \frac{1}{6}, \frac{5}{6}, 1\}$

Borel Sigma-Algebra

- $\mathcal{B}(\mathbb{R})$ is the smallest σ-algebra containing all open intervals of $\mathbb{R}$
- Includes all open/closed/half-open intervals, countable unions, intersections, complements
- It is the smallest σ-algebra containing sets of the form $(-\infty, x], \forall x \in \mathbb{R}$.

Remarks:

- Ω is called a sure event
- Any $A \in \mathcal{F}$ s.t. $\mathbb{P}(A) = 1$ is an almost-sure event
- $\mathbb{P}(A) = 1 \not\Rightarrow A = \Omega$
- $\mathbb{P}(A) = 0 \not\Rightarrow A = \emptyset$

Conditional Probability

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Given an event $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$, the **conditional probability** of an event $A \in \mathcal{F}$, is $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- **Chain/Multiplication Rule**:
  $\mathbb{P}(\cap_{i=1}^{n} A_i) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_1 \cap A_2) \cdots \mathbb{P}(A_n| \cap_{i=1}^{n-1} A_i)$
- **Total Probability Theorem**: Let $A_1, A_2, \cdots, A_n$ be disjoint events that partition the sample space. Assume that $\mathbb{P}(A_i) > 0$. Thus, for any $B \in \mathcal{F}$, we have $\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(A_i)\mathbb{P}(B|A_i)$
- **Bayes' Rule**: $\mathbb{P}(A_i|B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B|A_i)}{\sum_{i=1}^{n} \mathbb{P}(A_i)\mathbb{P}(B|A_i)}$

Independence

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Events A, $B \in \mathcal{F}$ are **independent** if
  - $\mathbb{P}(A|B) = \mathbb{P}(A)$ (can be used if $\mathbb{P}(B) > 0$) (OR)
  - $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ (can be used even when $\mathbb{P}(B) = 0$)
- Given an event $C \in \mathcal{F}$ with $\mathbb{P}(C) > 0$, A and B are **conditionally independent** if $\mathbb{P}(A \cap B|C) = \mathbb{P}(A|C)\mathbb{P}(B|C)$
- Independence does not imply conditional independence and vice-versa.

Random Variables

Fix $(\Omega, \mathcal{F})$.

A random variable $X : \Omega \to \mathbb{R}$ is a **measurable function**, i.e.,

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad X^{-1}(B) = \{\omega : X(\omega) \in B\} \in \mathcal{F}$$

Equivalently, a mapping $X : \Omega \to \mathbb{R}$ is a random variable if

$$\forall x \in \mathbb{R}, \quad X^{-1}((-\infty, x])) = \{\omega : X(\omega) \leq x\} \in \mathcal{F}$$

Example: Not a Random Variable

Let $\Omega = \{1,2,3,4,5,6\}$, $\mathcal{F} = \{\emptyset, \{1,2,3\}, \{4,5,6\}, \Omega\}$.

Define $X(\omega)$ as

$$X(\omega) = \begin{cases} 1, & \omega \in \{1,2,3\} \\ 2, & \omega \in \{4\} \\ 3, & \omega \in \{5,6\} \end{cases}$$

Then X is not a random variable since $X^{-1}\big((-\infty, 2.5]\big) = \{1,2,3,4\} \notin \mathcal{F}$.

Question: For the given $\mathcal{F}$, which functions are RVs?

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Given a random variable $X : \Omega \to \mathbb{R}$, its **cumulative distribution function** (CDF), $F_X : \mathbb{R} \to [0, 1]$, is defined as

$$F_X(x) = \mathbb{P}((-\infty, x]), \quad x \in \mathbb{R}$$

,

Properties of CDF:

1. $\lim_{x \to -\infty} F_x(x) = 0$, $\lim_{x \to \infty} F_x(x) = 1$
2. **Monotonicity** If $a \leq b$, then $F_X(a) \leq F_X(b)$
3. **Right-Continuity** F_X is right-continuous, i.e., for all $x \in \mathbb{R}$, $\lim_{\epsilon \downarrow 0} F_x(x + \epsilon) = F_X(x)$.

Discrete vs Continuous Random Variables

Discrete RV:

- Takes countable values
- Described by **Probability Mass Function (PMF)**:

$$p_X(x) = \mathbb{P}(X = x)$$

- Common Distributions: Bernoulli, Binomial, Geometric, Uniform, Poisson

Continuous RV:

- Takes values in intervals of $\mathbb{R}$
- Described by **Probability Density Function (PDF)**:

$$f_X(x) \; s.t. \; \mathbb{P}_X(a \leq X \leq b) = \int_a^b f_X(x)dx$$

- Common Distributions: Uniform, Exponential, Gaussian

PMFs of Common Distributions

- **Bernoulli** Ber(p): $p_X(x) = \begin{cases} p, & x = 1, \\ 1-p, & x = 0, \end{cases} \quad x \in \{0, 1\}.$
- **Binomial** Bin(n, p):
$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$
- **Geometric** Geo(p):
$$p_X(k) = (1-p)^{k-1} p, \quad k = 1, 2, 3, \ldots.$$
- **Discrete Uniform** Unif($\{a, \ldots, b\}$):
$$p_X(k) = \frac{1}{b-a+1}, \quad k = a, a+1, \ldots, b.$$
- **Poisson** Pos(λ):
$$p_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \ldots.$$

PDFs of Common Continuous Distributions

- **Uniform** $U[a, b]$:

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

- **Exponential** $\mathrm{Exp}(\lambda)$:

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

- **Gaussian (Normal)** $N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

Joint Distribution of Two Random Variables

Let X and Y be RVs associated with the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Joint CDF**: $F_{X,Y}(x,y) = \mathbb{P}(X \leq x, Y \leq y)$
- If X, Y discrete: **Joint PMF**: $p_{X,Y}(x,y) = \mathbb{P}(X = x, Y = y)$
- **Marginal PMFs**: $p_X(x) = \sum_y p_{X,Y}(x,y)$, $p_Y(y) = \sum_x {}_{X,Y}(x,y)$
- If X, Y jointly continuous: **Joint PDF**: $f_{X,Y}(x,y)$
  For $B \in \mathcal{B}(\mathbb{R})$, $\mathbb{P}((X,Y) \in B) = \int\int_B f_{X,Y}(x,y)dxdy$
- **Marginal PDFs**: $f_X(x) = \int_y f_{X,Y}(x,y)dy, \quad f_Y(y) = \int_x f_{X,Y}(x,y)dx$

Expectation

- Expectation: $\mathbb{E}[X]$
  - Discrete RV: $\mathbb{E}[X] = \sum x \cdot p_X(x)$
  - Continuous RV: $\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f_X(x)dx$
- **Linearity of Expectation**: $\mathbb{E}(\sum_{i=1}^{n} a_i X_i) = \sum_{i=1}^{n} a_i \mathbb{E}(X_i)$
- **Fundamental Theorem of Expectation**:
  - $\mathbb{E}[g(X)] = \sum g(x) \cdot p_X(x)$
  - $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x)dx$
- **Tail Probability Formula** for Expectation of a Continuous RV:

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X > x)dx - \int_0^{\infty} \mathbb{P}(X < -x)dx$$

Conditioning and Independence

Let X and Y be RVs associated with the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Conditional PMF**: $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$ for some y s.t $p_Y(y) > 0$
- **Conditional Expectation**: $\mathbb{E}(X|Y = y) = \sum_x x \cdot p_{X|Y}(x|y)$
- **Total Expectation theorem**: $\mathbb{E}(X) = \sum_y \mathbb{E}(X|Y = y) \cdot p_Y(y)$
- **Law of Iterated Expectations**: $\mathbb{E}(\mathbb{E}(X|Y)) = \mathbb{E}(X)$
- **Independence**: $p_{X,Y}(x, y) = p_X(x)p_Y(y), \forall (x, y)$
- **Conditional PDF**: $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$ for any y s.t $f_Y(y) > 0$
- If $X \perp Y$, $\mathbb{E}[XY] = \mathbb{E}(X)\mathbb{E}(Y)$

Moments

Consider a RV X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

- n-th Moment: $\mathbb{E}[X^n]$
- n-th Central Moment: $\mathbb{E}[(X - \mathbb{E}[X])^n]$
- Second central moment is called **variance**.
- **Variance**: $\boxed{\text{var(X)} = \text{E[(X - E[X])}^2\text{]}}$
- Standard Deviation of X : $\sqrt{var(X)}$
- For RVs X, Y defined on $(\Omega, \mathcal{F}, \mathbb{P})$,
    - Conditional Variance: $var(X|Y) = \mathbb{E}\left[\left(X - \mathbb{E}(X|Y)\right)^2 |Y\right]$
    - **Law of Total Variance**: $var(X) = \mathbb{E}\left[var(X|Y)\right] + var\left(\mathbb{E}(X|Y)\right)$
    - If $X \perp Y$, $var(X + Y) = var(X) + var(Y)$

Covariance and Correlation

Let X and Y be RVs associated with the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Covariance**: $cov(X, Y) = \mathbb{E}\big[(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))\big]$
- Alternatively: $cov(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$
- $\boxed{\text{cov(X,Y)=0} \Rightarrow X \text{ and } Y \text{ are uncorrelated.}}$
- **Independence** $\Rightarrow$ **Uncorrelatedness**, converse not true, in general
- **Correlation Coefficient**: $\rho(X, Y) = \frac{cov(X,Y)}{\sqrt{var(X)var(Y)}}$
- $\boxed{\text{-1} \leq \rho \leq 1}$ (Proof by Cauchy-Schwarz Inequality)

Moment Generating Functions

For a RV X associated with the probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **Moment Generating Function**,
  $$M_X(s) = \mathbb{E}[e^{sX}] = \begin{cases} \sum_x e^{sx} p_X(x), & X \text{ discrete} \\ \int_x e^{sx} f_X(x) dx, & X \text{ continuous} \end{cases}$$
- Associated **Region of Convergence** (ROC): $\{s : |M_X(s) < \infty\}$
- MGF of a continuous RV: Sign reversal of bilateral Laplace Transform
- Moment Generating Properties:
  $M_X(0) = 1, \; \frac{d}{ds} M_X(s)|_{s=0} = \mathbb{E}(X), \; \frac{d^n}{ds^n} M_X(s)|_{s=0} = \mathbb{E}(X^n)$
- **Linear function**: If $Y = aX + b$, then $M_Y(s) = e^{sb} M_X(as)$
- **Sum of independent RVs**: If $X \perp Y$, then
  $M_{X+Y}(s) = M_X(s) M_Y(s)$

Sum of a Random Number of Independent RVs

- Consider $X_1, X_2, ...$ i.i.d RVs with mean $\mathbb{E}(X)$ and variance $var(X)$
- Another RV N that takes non-negative values
- Let $Y = X_1 + X_2 + \cdots + X_N$
  - $\mathbb{E}(Y) = \mathbb{E}(N)\mathbb{E}(X)$
  - $var(Y) = \mathbb{E}(N)var(X) + (\mathbb{E}(X))^2 var(N)$
  - $M_Y(s) = M_N(\log(M_X(s)))$

Concentration Inequalities

- Useful for bounding tail probabilities
- **Markov Inequality:** For a RV $X \geq 0$ with $\mathbb{E}(X) < \infty$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0$$

- **Chebyshev Inequality:** For a RV X with mean μ and variance σ^2,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

- **Chernoff Bound:** Exponentially decaying bound

$$P(X \geq a) \leq \inf_{\substack{s>0 \\ s\in ROC}} \frac{\mathbb{E}[e^{sX}]}{e^{sa}}, \quad a > 0$$

Convergence of a Sequence of Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

$\{X_n\}_{n=1}^{\infty}$ be a **sequence of RVs** defined w.r.t $\mathcal{F}$.

Let X be another RV, also defined w.r.t $\mathcal{F}$.

- **Point-wise Convergence**:

$$X_n \xrightarrow{pointwise} X \iff \lim_{n\to\infty} X_n(\omega) = X(\omega)\}, \ \forall \omega \in \Omega$$

- **Almost-sure Convergence**:

$$X_n \xrightarrow{a.s.} X \iff \mathbb{P}(\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1$$

- **Convergence in Probability**:

$$X_n \xrightarrow{p} X \iff \forall \epsilon > 0, \ \lim_{n\to\infty} \mathbb{P}(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) = 0$$

- **Mean-Square Convergence**:

$$X_n \xrightarrow{m.s.} X \text{ if } \mathbb{E}[X_n^2] < \infty, \ \forall n, \text{ and } \lim_{n\to\infty} \mathbb{E}[(X_n - X)^2] = 0$$

- **Convergence in Distribution**: A sequence of RVs $\{X_n\}_{n=1}^{\infty}$ converges to a RV X in distribution,

$$X_n \xrightarrow{d} X \iff F_{X_n}(x) \to F_X(x) \text{ at all continuity points } x \text{ of } F_X$$

- Implications of various forms of convergence

$$\boxed{X_n \xrightarrow{a.s.} X \Rightarrow X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X}$$

$$\boxed{X_n \xrightarrow{m.s.} X \Rightarrow X_n \xrightarrow{p} X}$$

If $X_n \xrightarrow{d} c$, where c is a constant, then $X_n \xrightarrow{p} c$

Limit Theorems

Let $X_1, X_2, \ldots$ i.i.d. with mean μ, variance σ^2

- **Weak Law of Large Numbers** (WLLN):

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{p} \mu$$

- **Strong Law of Large Numbers** (SLLN):

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{a.s.} \mu$$

- **Central Limit Theorem** (CLT):

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n} (X_i - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

References

1. D. P. Bertsekas, J. N. Tsitsiklis, "Introduction to Probability", 2/e, Athena Scientific.
2. Bruce Hajek, "Random Processes For Engineers", Cambridge University Press, 2013.